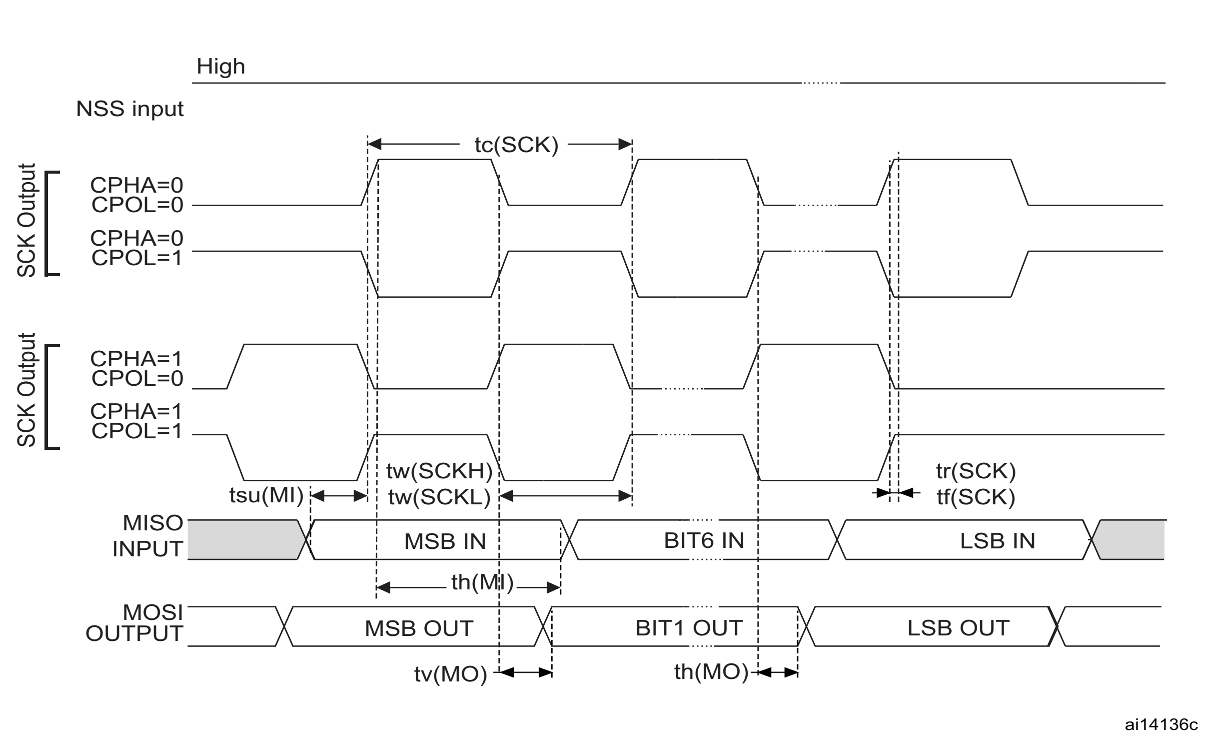
Figure 29. SPI timing diagram - master mode



1. Measurement points are done at 0.5 V_{DD} and with external C_L = 30 pF.

Table 65. I²S characteristics

Symbol	Parameter(1)	Conditions	Min	Max	Unit
f_{MCK}	I2S main clock output	-	-	48	MHz
f_{CK}	I2S clock frequency	Master TX	-	12	MHz
		Master RX	-	12	
		Slave TX	-	15	
		Slave RX	-	48	
$t_{v(WS)}$	WS valid time	Master mode	-	5	ns
$t_{h(WS)}$	WS hold time	Master mode	0	-	ns
$t_{su(WS)}$	WS setup time	Slave mode	3.5	-	ns
$t_{h(WS)}$	WS hold time	Slave mode	1.5	-	ns
$t_{su(SD\_MR)}$	Data input setup time	Master receiver	5	-	ns
$t_{su(SD\_SR)}$		Slave receiver	2.5	-	ns
$t_{h(SD\_MR)}$	Data input hold time	Master receiver	2.5	-	ns
$t_{h(SD\_SR)}$		Slave receiver	1	-	ns
$t_{v(SD\_ST)}$	Data output valid time	Slave transmitter (after enable edge)	-	19.5	ns
$t_{v(SD\_MT)}$		Master transmitter (after enable edge)	-	5	ns